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Original Article

A formalin-fixed paraffin-embedded (FFPE)-based prognostic signature to predict metastasis in clinically low risk stage I/II microsatellite stable colorectal cancer

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ABSTRACT

Approximately 20% early-stage (I/II) colorectal cancer (CRC) patients develop metastases despite curative surgery. We aim to develop a formalin-fixed and paraffin-embedded (FFPE)-based predictor of metastases in early-stage, clinically-defined low risk, microsatellite-stable (MSS) CRC patients. We considered genome-wide mRNA and miRNA expression and mutation status of 20 genes assayed in 150 fresh-frozen tumours with known metastasis status. We selected 193 genes for further analysis using NanoString nCounter arrays on corresponding FFPE tumours. Neither mutation status nor miRNA expression improved the estimated prediction. The final predictor, ColoMet19, based on the top 19 genes' mRNA levels trained by Random Forest machine-learning strategy, had an estimated positive-predictive-value (PPV) of 0.66. We tested ColoMet19 on an independent test-set of 131 tumours and obtained a population-adjusted PPV of 0.67 indicating that early-stage CRC patients who tested positive have a 67% risk of developing metastases, substantially higher than the metastasis risk of 40% for node-positive (Stage III) patients who are generally treated with chemotherapy. Predicted-positive patients also had poorer metastasis-free survival (hazard ratios [HR] = 1.92, design-set; HR = 2.05, test-set). Thus, early-stage CRC patients who test positive may be considered for adjuvant therapy after surgery.

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Abbreviations: CRC, colorectal cancer; FFPE, formalin-fixed and paraffin-embedded; MSS, microsatellite stable; HR, hazard ratio; PPV, Positive Predictive Value; NPV, Negative Predictive Value; FAP, familial adenomatous polyposis; RMA, robust multi-chip average; FLOO, full-leave-one-out; OOB, out-of-bag; ROC, receiver operating characteristic; TP, true positive; FP, false positive; FN, false negative; TN, true negative; LN, lymph node; CI, confidence interval.

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Introduction

Colorectal Cancer (CRC) is the third most common cancer globally [1]. 40% of patients have stage I or II localized disease at diagnosis, a proportion that could increase as screening becomes more prevalent. The optimal management of early-stage CRC can be controversial [2,3]. Most patients are cured by surgery alone; however, approximately 20% develop systemic metastases, which are often incurable [4]. Thus, it is important to identify biomarkers that can risk-stratify patients with early-stage CRC. Accurate prediction of early-stage CRC likely to metastasize will identify patients who may benefit from adjuvant chemotherapy. Conversely, patients predicted to remain disease-free would be spared unnecessary costs and the adverse effects of adjuvant therapy.

Current gene signatures to estimate metastasis risk in early-stage CRC have four major limitations. First, most signatures such as ColoPrint [5] are suitable only for frozen tumour samples, which, unlike formalin-fixed, paraffin-embedded (FFPE) samples, are not routinely available in clinical practice. Second, currently available test for FFPE samples, Oncotype DX, has rather low discriminative value for stage II patients and hence limited clinical utility. A recent retrospective analysis found that Stage II patients with the highest recurrent risk-scores had a 19% risk of metastases compared to 9% for patients with the lowest risk-scores [6]. Interestingly, another recent study in early-stage breast cancer has shown that a second generation assay was able to significantly re-stratify patients with low and intermediate risk by the Oncotype Recurrence Score into clinically relevant different risk groups thus demonstrating increased prognostic accuracy over this first generation assay [7]. Third, these signatures were designed and validated in heterogeneous patient populations. A recent study found that when evaluated in an independent patient cohort, current signatures only marginally improve risk estimates over those based on established clinical factors [8]. Fourth, for signatures to be used as decision aids in clinical practice, methods to assign risk prospectively to individual patient tumours profiled one sample at-a-time, rather than methods that depend upon normalization within patient cohorts, are required [9].

To address these issues, we sought to design a prognostic signature for the very group of patients where prediction of metastasis risk would be most useful, namely stage I/II patients who would normally be treated by surgery alone but are nevertheless susceptible to metastasis. We aimed to design a predictor based on readily-available FFPE samples. We considered three types of data. (1) mRNA expression data, as measured by NanoString nCounter assays which are especially suited to the highly fragmented mRNA present in FFPE samples. (2) miRNA expression levels, some of which may be prognostic [10–12]. (3) Somatic mutations in 20 frequently mutated genes in CRC, as these may be prognostic [13]. We sought to achieve a high positive predictive value (PPV) for metastasis so that the test could be used to select patients that might benefit from adjuvant chemotherapy. We judged that a useful predictor would have a PPV >0.4, which is the metastasis risk for patients with stage III tumours, for whom adjuvant chemotherapy is generally recommended [14,15].

Materials and methods

Fig. 1 describes the overall design of the study, which adhered to a strict separation of the design of the predictor from its testing and evaluation.

Patients and tumour samples

For the first phase of predictor design, we began with a total of 150 frozen tumour samples (Fig. 1); expression data from 70 of these patients were generated as part of a previous study [16], and 80 were newly selected and expression profiled. All were patients of Chinese ancestry with stage I or II colorectal cancer, who underwent surgery at Singapore General Hospital between 2001 and 2012. We excluded patients with any of following characteristics: diagnosis of familial adenomatous polyposis (FAP) or hereditary non-polyposis CRC; $\geq$ 3 polyps in the surgically resected mucosa possibly affected by undiagnosed FAP or another familial syndrome; age < 50 years, and other factors associated with increased risk of metastasis including T4 lesions, intestinal perforation during surgery, or poor lymph-node harvest. We also excluded patients with microsatellite instable tumours, because these are a small subset of sporadic CRCs with different biology from microsatellite stable CRC [17]. To further remove additional sources of variation, we focused on left-sided tumours (to the left of splenic flexure), as left and right-sided tumours are reported to have different biology [18]. We determined the metastasis-positive patients from their clinical records. The metastasis-negative patients were metastasis-free for $\geq$5 years of follow-up at the time of selection, with the exception of seven patients from a previous study [16] who had less than 5 years of follow-up. Survival was determined from the Singapore National Registry of Disease database. Table 1 and Supplementary Table S1 provide detailed patient information. There was no significant difference in the clinicopathological features of the previously published (n = 70) and the newly recruited (n = 80) fresh-frozen samples in all the parameters examined except for the small difference in the number of lymph nodes examined for metastasis-negative patients (Supplementary Table S1). For the second phase of predictor design we obtained matched FFPE samples from 144 of the 150 fresh-frozen tumours studied; FFPE samples for the remaining 6 were unavailable or did not yield usable RNA.

For testing the predictor, we obtained FFPE samples from 94 metastasis-negative and 37 metastasis-positive tumours from the Singapore General Hospital Department of Pathology. These were from independent patients (non-overlapping with the first cohort) diagnosed between 2001 and 2013, and the inclusion and exclusion criteria were as for tumours used to design the predictor.

The only significant difference in clinico-pathological characteristics between the tumours used in designing the predictor and those used for testing it was a median age of 66 in the design set and 64 in the test set (uncorrected p = 0.04; Table 1, Supplementary Table S2).

Microarray gene expression profiling and analysis

The processing of the first 70 frozen tumours for mRNA profiling was as reported [16]. We macro-dissected the remaining 80 samples as previously performed to enrich for tumour cells, then extracted RNA with Qiagen RNeasy mini kit (Qiagen, Hilden, Germany). We synthesized cDNA from 100 ng/μl of total RNA using Affymetrix (Santa Clara, CA) IVT-expression kit protocol. All 150 frozen tumours were assayed with the Affymetrix U133_Plus2 array. Probe set expression levels were normalized using the robust multi-chip average (RMA) approach [19,20]. Batch effects were removed using the R-package ComBat [21,22], and the resulting expression values were analyzed in Partek Genomic Suite™ version 6.6, (Partek Inc., St. Louis, MO) with 3-way ANOVA and a full-leave-one-out (FLOO) two-level nested cross-validation. The independent variables were metastasis status, and, to exclude possible confounds, gender and scan date. The FLOO two-level nested cross-validation consisted of an "inner" cross validation to identify the optimal predictors and the best classifier for discrimination of metastasis status, and an "outer" cross-validation to obtain an unbiased estimation of accuracy [23].

NanoString gene expression

We considered the potentially informative probe sets from the Partek analysis together with probe sets identified previously [16]. NanoString (NanoString Technologies Inc., Seattle, WA) designed custom CodeSets (i.e. hybridization probes) corresponding to the Affymetrix probe sets. These CodeSets were assayed on RNA from 97 metastasis-negative and 47 metastasis-positive FFPE matched samples of the fresh-frozen tumours.

For nCounter profiling, a 10 μm thick section of each FFPE sample was deparaffinized with xylene, and total RNA was extracted using the Roche® High Pure FFPET RNA Isolation kit version 3 (Roche, Mannheim, Germany) according to the manufacturer's instructions. 200 ng of total RNA was hybridized with the NanoString CodeSets. Raw counts of each target gene were normalized by the geometric mean counts of 11 housekeeping genes (*HRNP1, RPL27, RPL9, RPL6, RPL30, OAZ1, PTMA, RPS29, UBC, RPS12* and *RPS16*) and spiked-in controls. The normalized data were $\log_2$ transformed for further analysis.

miRNA profiling

We used the mirVana® miRNA isolation Kit (Life Technologies, CA, USA) to extract total RNA from each of the 148 frozen tumours used in designing the predictor. Remaining 2 frozen tumours were unavailable. 100 ng of total RNA was hybridized onto the Agilent Human miRNA microarray release 21.0 8 × 60 K (Agilent Technologies Inc., Santa Clara, CA) according to the manufacturer's protocol. We carried out quantile normalization with each of 13 batches, measurements <1 were set to 1, and the data were $\log_2$ transformed. Probes with more than 50% of missing values across the samples were excluded from further analysis. We used ComBat to remove batch effects from the combined data [21,22].

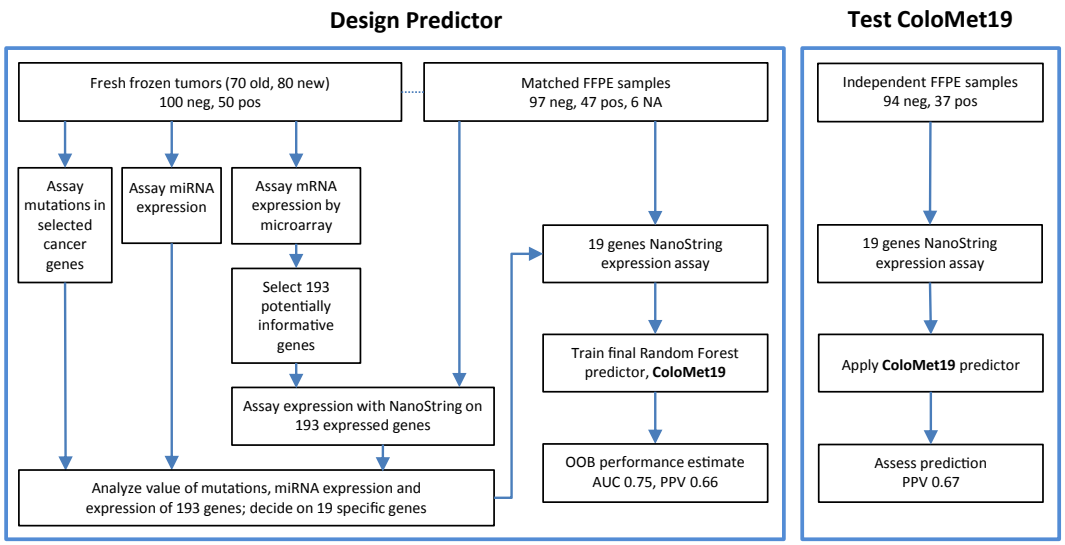

Fig. 1. Study Design. The study consisted of a design phase (left), in which ColoMet19 was designed, and a strictly independent test phase (right). In the design phase (left), we first measured the expression levels of a large number of genes using microarrays and RNA from fresh-frozen tumour samples. We selected 193 genes, for which we designed NanoString nCounter assays that we applied to FFPE tumour samples corresponding to the original frozen tumour samples. We also considered miRNA expression and mutations in selected cancer-related genes. However, including these data did not improve prediction. We then selected 19 of the 193 genes for expression analysis with nCounter assays plus a Random Forest voting threshold, which together constitute the ColoMet19 predictor. We tested ColoMet19 on an independent set of FFPE tumour samples (right).

Table 1
Clinico-pathological characteristics of FFPE tumour samples.

Characteristics	Design data[a] (n = 144)				p	Independent test data (n = 131)				p	p[b]
	Metastasis positive (n = 47)		Metastasis negative (n = 97)			Metastasis positive (n = 37)		Metastasis negative (n = 94)			
	No. of patients	%	No. of patients	%		No. of patients	%	No. of patients	%		
Age, years											
Median	67		66		0.44	64		64.5		0.42	0.04
≤70	35	74.5	63	64.9	0.34	28	75.7	68	72.3	0.83	0.36
>70	12	25.5	34	35.1		9	24.3	26	27.7		
Gender					1.00					0.17	0.63
Male	24	51.1	51	52.6		22	59.5	42	44.7		
Female	23	48.9	46	47.4		15	40.5	52	55.3		
Localisation					0.47					0.52	0.31
Left Colon	28	59.6	64	66		28	75.7	64	68.1		
Rectum	19	40.4	33	34		9	24.3	30	31.9		
Tumour differentiation					0.45					0.63	0.83
Well	7	14.9	12	12.4		5	13.5	18	19.1		
Moderate	39	83	83	85.6		31	83.8	74	78.7		
Poor	1	2.1	0	0		0	0	1	1.1		
Others	0	0	2	2		1	2.7	1	1.1		
No. of LN examined					0.48					0.43	1.00
10-12	9	19.1	14	14.4		4	10.8	17	18.1		
>12	38	80.9	83	85.6		33	89.2	77	81.9		
Metastasis site					NA					NA	0.81
Liver	8	17	NA	NA		6	16.2	NA	NA		
Lung	11	23.4	NA	NA		6	16.2	NA	NA		
Recurrent	8	17	NA	NA		10	27	NA	NA		
Multiple	17	36.2	NA	NA		12	32.4	NA	NA		
Other	3	6.4	NA	NA		3	8.1	NA	NA		
Chemotherapy					2e-7					2e-6	0.39
Yes	16[c]	34.0	2	2.1		16[c]	43.2	6	6.4		
No	31	66.0	95	97.9		21	56.8	88	93.6		
Stage					0.16					0.05	0.11
I	5	10.6	21	21.6		5	13.5	30	29.8		
II	42	89.4	76	78.4		32	86.5	64	70.2		

Notes: LN: lymph node; No: number; NA: not applicable.
[a] FFPE tumour samples corresponded to the fresh frozen tumour samples in Table S1.
[b] p-value for statistical tests between design data and independent test data.
[c] Palliative chemotherapy for metastasis positive patients.

Targeted sequencing for mutation detection

Genomic DNA (gDNA) from 150 tumours and 150 matched mucosa was extracted using a high-salt protocol [24]. PCR primers targeting hotspots of 20 commonly-mutated genes (Supplementary Table S3) were designed by Fluidigm (Fluidigm, South San Francisco, CA). Amplicon libraries were generated from 50 ng/µl of gDNA template and pooled using the Fluidigm Access Array system according to the manufacturer's protocols. Amplicons were sequenced as 250 or 300 paired-end reads on Illumina (San Diego, CA) MiSeq sequencers using the MiSeq Reagent Kit v3. Mean coverage ranged from 350–450×. Somatic variants were called using either GATK (version 1.0; https://www.broadinstitute.org/gatk/) with tumour-minus normal subtraction or MuTect (version 1.1.4; https://www.broadinstitute.org/cancer/cga/mutect).

Statistics and Random Forest implementation

We used the Random Forest [25] implementation from the Python machine learning package *scikit-learn*, v 0.17 as the machine-learning framework for the predictor [26]. Random Forest implementation uses training data to derive the "out-of-bag" (OOB) estimate of performance in new data which serves the same purpose as those obtained by cross validation [27]. Random Forests also rank features according to their importance in prediction. Survival analyses (Kaplan-Meier, Cox proportional hazards) were performed in the Python *lifelines* package [28]. Additional statistical tests were performed in Python using SciPy package [29] (Mann-Whitley, Fisher's exact test) and the R statistical programming environment (Fisher's exact test).

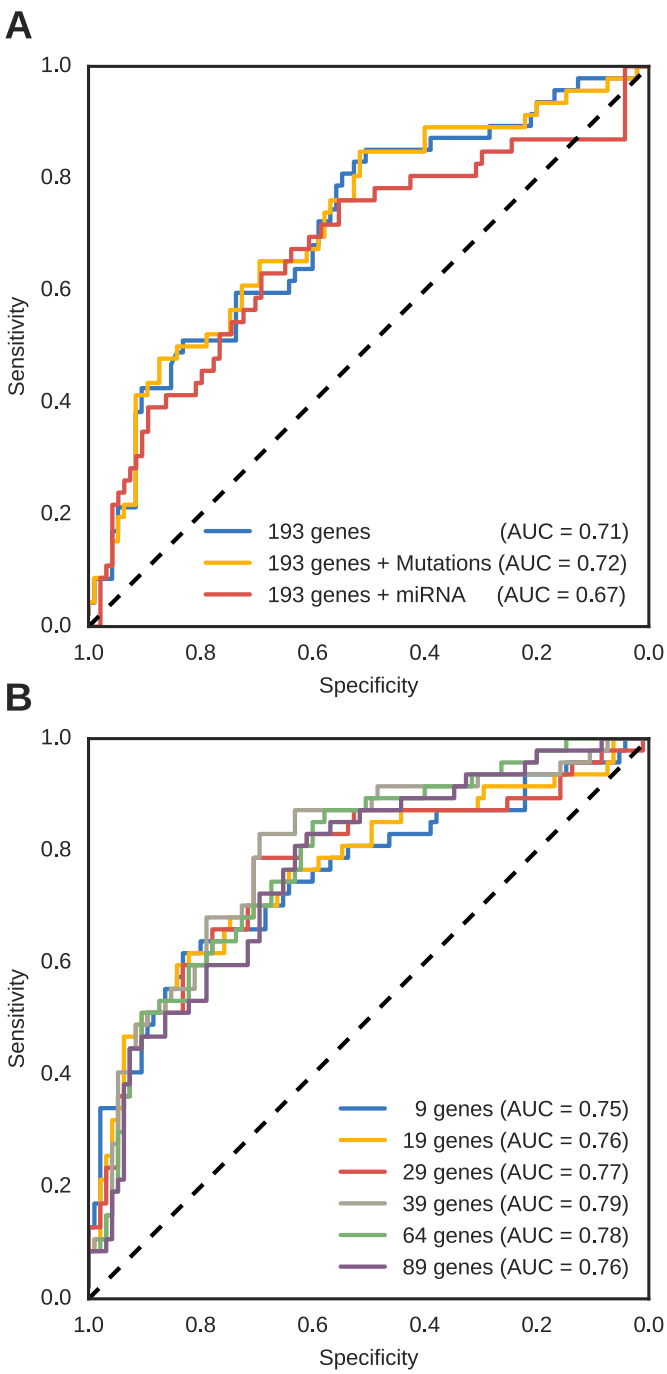

Fig. 2. Receiver operating characteristic (ROC) curves for Random Forest feature selection. (A) ROC curves for predictors with the expression levels of 193 genes alone (blue), 193 genes as assayed by NanoString nCounter and mutations (yellow), and 193 genes and miRNA levels (red). (B) ROC curves for predictors using the expression levels of different numbers of genes as assayed by nCounter.

Results

Our goal was to develop an FFPE-based predictor of metastasis of left-sided, MSS, stage I and II colorectal cancers. We targeted a PPV substantially >0.4, a metastasis rate observed for node-positive stage III colorectal cancer [14,15] for which adjuvant therapy is recommended to reduce metastasis risk. We enforced a strict separation between data used for designing and testing the predictor (Fig. 1). We started with microarray expression data on 50 metastasis-positive and 100 metastasis-negative frozen tumour samples to select genes. Out of approximately 54,000 probe sets, we identified 258 probe sets as discriminative regarding metastases.136 of these were up-regulated in metastasis-positive tumours; the remaining 122 were down-regulated. The 258 probe sets corresponded to 201 genes and were submitted to NanoString for CodeSet design. It was possible to design CodeSets for 193 of the 201 genes.

We used nCounter to measure the mRNA levels of the 193 genes in FFPE samples corresponding to 144 of the 150 frozen tumour samples. The nCounter measurements were reasonably well correlated with the microarray measurements (Pearson R = 0.68, Supplementary Fig. S1), especially for genes with Affymetrix RMA expression levels >5 (R = 0.77).

Design of ColoMet19

All Random Forests consisted of 100,000 decision trees, each trained on an independent subsample of the data drawn with replacement. Predictions are made by voting across the trees, and the voting threshold for a positive prediction is adjusted to achieve the desired trade-offs between specificity and sensitivity and between PPV and negative predictive value (NPV).

Design of the final predictor proceeded in two stages. First, we assessed whether including miRNA expression levels would improve prediction. No miRNA showed significant differential expression between the metastasis-positive and -negative tumours after false discovery rate correction, and inclusion of miRNA

Table 2
Confusion matrix of the 19-gene signature (population adjusted PPV and NPV).

Real\predicted	Design set		Test set	
	Metastasis positive	Metastasis negative	Metastasis positive	Metastasis negative
Metastasis positive	TP: 19	FN: 28	TP: 13	FN:24
Metastasis negative	FP: 5	TN: 92	FP: 4	TN: 90
	PPV: 0.79 (0.66)	NPV: 0.77 (0.86)	PPV: 0.76 (0.67)	NPV: 0.79 (0.86)

PPV: positive predictive value; NPV: negative predictive value; TP: true positive; FN; false negative; FP: false positive, TN: true negative.

expression levels along with the 193 gene-expression levels did not improve prediction (Fig. 2A). Frequencies of somatic mutation were similar to those reported in The Cancer Genome Atlas Network [30] and included *APC* (81%), *TP53* (68%) and *KRAS* (33%). Only *KRAS* had significantly different numbers of mutations between metastasis-negative and -positive tumours (uncorrected $p = 0.015$). Likewise, including mutation data from these 20 genes did not improve prediction (Fig. 2A).

Second, having determined that mRNA expression levels alone supported prediction as accurate as that obtained by including miRNA expression or mutation status, we ranked the 193 genes as assayed by nCounter by training a Random Forest on them. We tested several sets of genes of different sizes, namely the top 9 genes as ranked by Random Forest, the top 19, the top 29, etc. The sizes of these sets were based on the most cost-efficient NanoString assay sizes (20, 30, 40, 50, 75, 100) while reserving space for 11

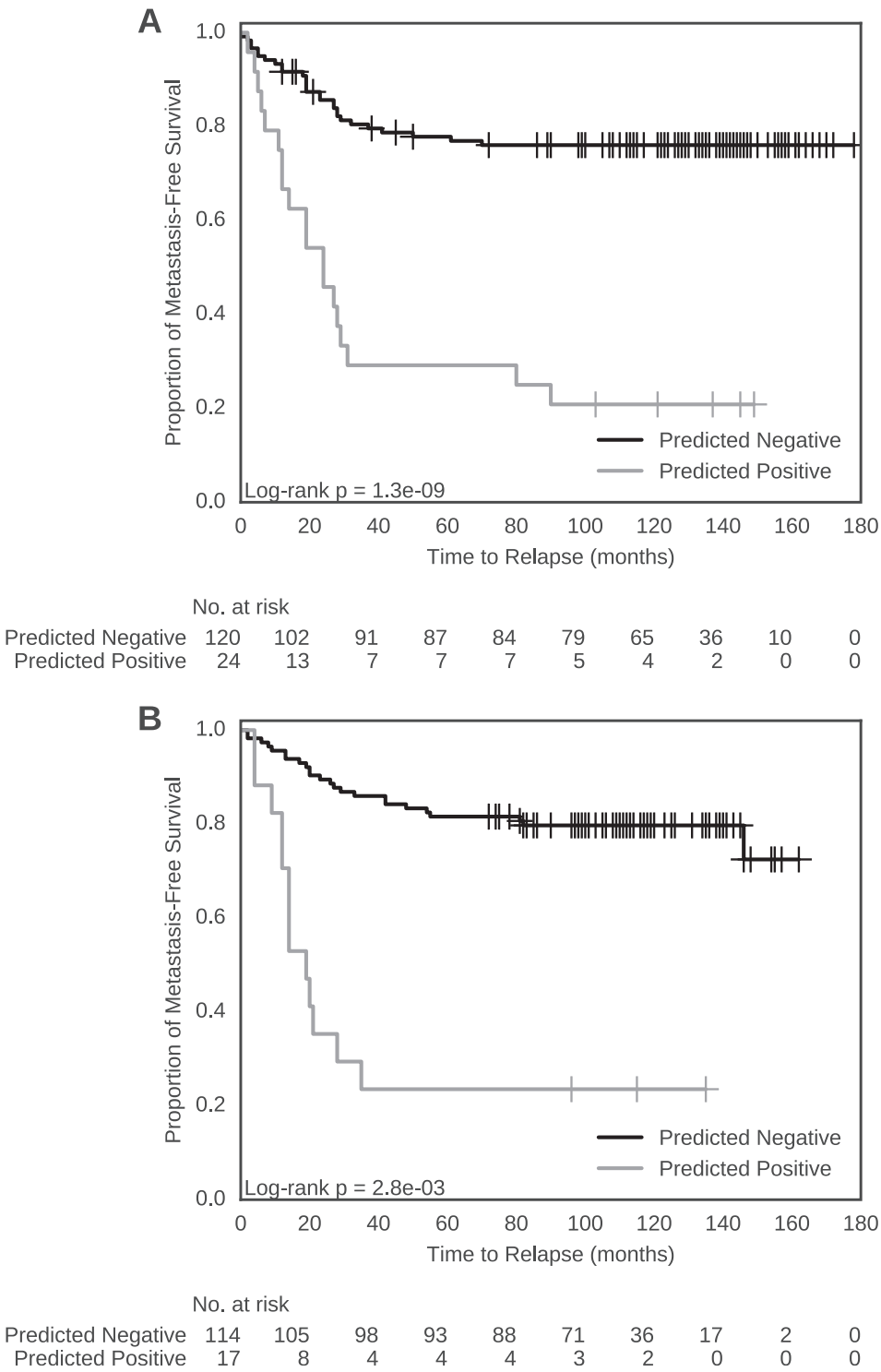

Fig. 3. Kaplan-Meier plots for metastasis-free survival in (A) design and (B) test data.

housekeeping genes. In all, we looked at gene sets of size 9, 19, 29, 39, 64, and 89. We evaluated each set of genes by training a new Random Forest. Examination of the receiver operating characteristic (ROC) curves based on OOB misclassification estimates and the areas under the ROC curves indicated that predictors using the 19 gene set were as accurate as the others (Fig. 2B). We selected these 19 genes as the basis for the predictor, which we termed ColoMet19. A voting threshold of 0.576 (Supplementary Fig. S2) was selected, which corresponded to a sensitivity of 0.4, a specificity of 0.95, a PPV of 0.66, and an NPV of 0.86 as assessed by OOB estimates given a 20% baseline metastasis risk (Table 2). With focus on clinical application, our selection criteria were high PPV and reasonable sensitivity; a PPV of 0.66 substantially exceeds our goal of a PPV >0.4, which is the metastasis risk for patients with stage III tumours, for whom adjuvant chemotherapy is generally recommended.

Evaluation of ColoMet19

We evaluated the predictive performance of the selected Random Forest predictor using the selected 19 genes (Supplementary Table S4) and voting threshold on an independent test set of 131 FFPE tumour samples. The mRNA expression levels of the 19 genes were generally consistent between the data used in design and testing of ColoMet19 (Supplementary Fig. S3). The classification results were comparable to those achieved on the design set, with a PPV in the test set of 0.67 and an NPV of 0.86 given a baseline metastasis risk of 20% (Table 2).

Consistent with the high PPV and NPV, the metastasis-free survival for predicted metastasis-positive and -negative status were significantly different in both the data used for design and for testing (Fig. 3). Multivariable Cox-proportional hazards analysis confirmed a substantial survival difference: the hazard ratio for predicted positive tumours compared to predicted negative tumours was 1.92 (95% CI 1.53 to 2.4) in the tumours used to design ColoMet19 and 2.05 (95% CI 1.6 to 2.62) in the test data (Table 3).

Discussion

We designed and validated ColoMet19, a 19 gene FFPE-based prognostic signature with high PPV for metastasis prediction in early-stage, MSS CRC (Table 2). The PPV of 0.67 in the ColoMet19 test data indicates that, compared to a baseline risk of 20%, early-stage MSS-CRC patients for which our assay tests positive would have a 67% probability to develop metastases. Thus, such patients have substantially greater risk of developing metastases than stage III CRC patients, for whom adjuvant therapy is generally recommended. This suggests that early-stage CRC patients who tested positive by the ColoMet19 predictor should also be considered for adjuvant therapy after surgery. Conversely, the NPV of 0.86 implies that a stage I/II CRC patient who tested negative would have an 86% chance of remaining metastasis free, and thus need not be subjected to the toxicity of further therapy. The discriminatory power achieved by ColoMet19 compares favourably with that reported in a recent retrospective gene expression risk-stratification study that included stage II colon cancers [6].

Although *KRAS* is found to be significantly differentially mutated in metastasis-positive patients suggesting its role in metastasis, *KRAS* mutation status did not add to the predictive value of the 19 expression-based genes. Neither did the miRNA profiles add value to the predictor. This indicates that dysregulated pathways leading to metastasis are adequately captured by gene expressions alone. This makes translation to the clinic easier since only one platform (NanoString nCounter) needs to be considered and hence made the test much more cost effective.

The 19 genes in the predictor encompass diverse cellular functions ranging from cell cycle control to cytokinesis, mRNA processing to helicase activity, protein folding to degradation, translation initiation, MUC-2, fatty acid and nucleic acid synthesis to chromatin remodeling, as well as voltage-gated potassium channel-related functions (Supplementary Table S4). Several of these genes are also involved in mitochondria dynamics such as importing of nuclear proteins, mitochondria tubular integrity and splicing. Contrary to recent studies, the inflammation and immune functions are not directly implicated in these genes. This does not imply that these functions are not important but rather perturbation in expressions of the 19 genes led to recurring disruption of underlying homeostasis that is probably the first-line barrier to metastasis.

There are other proposed methods to ascertain risk of metastasis in early-stage CRC. Several signatures have been published or even made commercially available [5], however, they may not add incremental value beyond established risk factors [8]. To our knowledge, ColoMet19 does not share genes with commercially available signatures such as Oncotype Dx Colon and ColoPrint. This may be due to uniqueness of our study design which focused on clinically low-risk cohorts, differences in the platforms and ethnicity of the patient populations. Nevertheless, several genes in ColoMet19 have been previously reported as useful in connection with CRC diagnosis and treatment, including SEPT9 [31] and AGR2 [32]. Immunohistochemistry for differentiation markers (*CDX2*) [33] and immune context of the tumour microenvironment (immunoscore) [34] may also be prognostic. The presence of circulating tumour cells [35] or DNA [36] after surgery may also indicate presence of minimal residual disease and elevated metastasis risk. Possibly, some of these indicators are complementary and could add to clinicians' ability to risk stratify stage I/II CRC patients for adjuvant therapy, and this possibility awaits further study.

In summary, we successfully designed and validated ColoMet19, an expression-based predictor suitable for use with FFPE samples that estimates the propensity for metastasis of early-stage low-risk MSS CRC patients. We were able to do this because of the unique combination of strict patient inclusion criteria, the power of genome-wide techniques, the availability of clinical data, and the robustness of bioinformatics analyses. ColoMet19 would need to be validated in prospective studies using the same voting threshold on independent sets of FFPE tumour samples from other centers before translation to clinical practice, and tested in other ethnicities to determine its breadth of applicability. Similarly, ColoMet19's utility for right-sided tumours or tumours in younger patients would need to be assessed. The adoption of this test could then improve the selection of early-stage CRC patients for adjuvant

Table 3
Multivariate cox proportional hazards analyses of predicted metastasis-positive versus −negative tumours.

Covariate	Design			Test		
	HR	95% CI	p	HR	95% CI	p
Age	1.02	0.77 to 1.35	0.89	1.07	0.77 to 1.50	0.68
Gender (female v male)	1.00	0.75 to 1.34	0.99	0.69	0.49 to 0.98	0.04
CA Location (colon v rectum)	0.90	0.67 to 1.20	0.47	1.17	0.81 to 1.67	0.40
Stage (II v I)	1.42	0.99 to 2.05	0.06	1.37	0.89 to 2.11	0.15
ColoMet19 prediction (positive v negative)	1.92	1.53 to 2.40	<0.0001	2.05	1.60 to 2.62	<0.0001

HR: hazard ratio; CI: confidence interval.

therapy, which could increase the quality adjusted life years of these patients while decreasing healthcare costs.

Availability of data and materials

Gene and miRNA microarray data, and gene nCounter data have been deposited at Gene Expression Omnibus (http://www.ncbi.nlm.nih.gov/geo/) under accessions GSE81986, GSE81980, GSE81981, GSE81983, and GSE81985. (link for review: http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?token=inmdikeutnqrlux&acc=GSE81986). Sequencing data have been deposited at the European Nucleotide Archive (http://www.ebi.ac.uk/ena) under accession PRJEB14177.

Disclosures

The authors will file an invention disclosure with National University of Singapore, and may seek patent protection for Colo-Met19 in order to facilitate its widespread availability.

Author contributions

IT, SR, PYC contributed to conception and study design; YSL, CB, SAT, YYC, JYSW, TKHL, CLT, MHC, PT, PYC collected and assembled the data; YSL, CB, JP, CC, IT, SR, PYC contributed to data analysis and interpretation; SR, PYC provided financial support; TKHL, CLT, MHC, PYC provided study material or patients and TKHL, CLT, MHC, SR, PYC provided administrative support. All participated in manuscript writing and approval of final draft.

Consent for publication

Not Applicable.

Ethics approval and consent to participate

This study was approved by the SingHealth Centralized Institutional Review Board (2013/234/B), and informed consent was taken from all recruited patients.

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Conflicts of interest

The authors declare that they have no competing interests.

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.canlet.2017.05.031.

References

[1] J. Ferlay, I. Soerjomataram, R. Dikshit, S. Eser, C. Mathers, M. Rebelo, et al., Cancer incidence and mortality worldwide: sources, methods and major patterns in GLOBOCAN 2012, Int. J. Cancer 136 (2015) E359–E386.
[2] E.S. O'Connor, D.Y. Greenblatt, N.K. LoConte, R.E. Gangnon, J.I. Liou, C.P. Heise, et al., Adjuvant chemotherapy for stage II colon cancer with poor prognostic features, J. Clin. Oncol. 29 (2011) 3381–3388.
[3] N.J. Meropol, Ongoing challenge of stage II colon cancer, J. Clin. Oncol. 29 (2011) 3346–3348.
[4] M.R. Weiser, R.G. Landmann, M.W. Kattan, M. Gonen, J. Shia, J. Chou, et al., Individualized prediction of colon cancer recurrence using a nomogram, J. Clin. Oncol. 26 (2008) 380–385.
[5] R. Salazar, P. Roepman, G. Capella, V. Moreno, I. Simon, C. Dreezen, et al., Gene expression signature to improve prognosis prediction of stage II and III colorectal cancer, J. Clin. Oncol. 29 (2011) 17–24.
[6] T. Yamanaka, E. Oki, K. Yamazaki, K. Yamaguchi, K. Muro, H. Uetake, et al., 12-Gene recurrence score assay stratifies the recurrence risk in stage II/III colon cancer with surgery alone: the SUNRISE study, J. Clin. Oncol. 34 (2016) 2906–2913.
[7] I. Sestak, Y. Zhang, B.E. Schroeder, C.A. Schnabel, M. Dowsett, J. Cuzick, et al., Cross-stratification and differential risk by breast cancer index and recurrence score in women with hormone receptor-positive lymph node-negative early-stage breast cancer, Clin. Cancer. Res. 22 (2016) 5043–5048.
[8] A.F. Di Narzo, S. Tejpar, S. Rossi, P. Yan, V. Popovici, P. Wirapati, et al., Test of four colon cancer risk-scores in formalin fixed paraffin embedded microarray gene expression data, J. Natl. Cancer. Inst. 106 (10) (2014) dju247.
[9] E.R. Paquet, M.T. Hallett, Absolute assignment of breast cancer intrinsic molecular subtype, J. Natl. Cancer. Inst. 107 (2015) 357.
[10] T.F. Hansen, S. Kjaer-Frifeldt, R.D. Christensen, S. Morgenthaler, T. Blondal, J. Lindebjerg, et al., Redefining high-risk patients with stage II colon cancer by risk index and microRNA-21: results from a population-based cohort, Br. J. Cancer 111 (2014) 1285–1292.
[11] M.L. Slattery, J.S. Herrick, L.E. Mullany, N. Valeri, J. Stevens, B.J. Caan, et al., An evaluation and replication of miRNAs with disease stage and colorectal cancer-specific mortality, Int. J. Cancer 137 (2015) 428–438.
[12] K. Hur, Y. Toiyama, A.J. Schetter, Y. Okugawa, C.C. Harris, C.R. Boland, et al., Identification of a metastasis-specific MicroRNA signature in human colorectal cancer, J. Natl. Cancer. Inst. 107 (2015).
[13] J. Yu, W.K. Wu, X. Li, J. He, X.X. Li, S.S. Ng, et al., Novel recurrently mutated genes and a prognostic mutation signature in colorectal cancer, Gut 64 (2015) 636–645.
[14] V.L. Tsikitis, D.L. Larson, B.G. Wolff, G. Kennedy, N. Diehl, R. Qin, et al., Survival in stage III colon cancer is independent of the total number of lymph nodes retrieved, J. Am. Coll. Surg. 208 (2009) 42–47.
[15] D. Sargent, A. Sobrero, A. Grothey, M.J. O'Connell, M. Buyse, T. Andre, et al., Evidence for cure by adjuvant therapy in colon cancer: observations based on individual patient data from 20,898 patients on 18 randomized trials, J. Clin. Oncol. 27 (2009) 872–877.
[16] Y. Hong, T. Downey, K.W. Eu, P.K. Koh, P.Y. Cheah, A 'metastasis-prone' signature for early-stage mismatch-repair proficient sporadic colorectal cancer patients and its implications for possible therapeutics, Clin. Exp. Metastasis 27 (2010) 83–90.
[17] S. Popat, R. Hubner, R.S. Houlston, Systematic review of microsatellite instability and colorectal cancer prognosis, J. Clin. Oncol. 23 (2005) 609–618.
[18] T.J. Price, C. Beeke, S. Ullah, R. Padbury, G. Maddern, D. Roder, et al., Does the primary site of colorectal cancer impact outcomes for patients with metastatic disease? Cancer 121 (2015) 830–835.
[19] Z. Wang, M.G. Lewis, M.E. Nau, A. Arnold, M.T. Vahey, Identification and utilization of inter-species conserved (ISC) probesets on Affymetrix human GeneChip platforms for the optimization of the assessment of expression patterns in non human primate (NHP) samples, BMC Bioinforma. 5 (2004) 165.
[20] R.A. Irizarry, B.M. Bolstad, F. Collin, L.M. Cope, B. Hobbs, T.P. Speed, Summaries of Affymetrix GeneChip probe level data, Nucleic Acids Res. 31 (2003) e15.
[21] J. Leek, W. Johnson, H. Parker, E. Fertig, A. Jaffe, J. Storey, sva: Surrogate Variable Analysis, R package version 3.18.0.ed, 2016.
[22] W.E. Johnson, C. Li, A. Rabinovic, Adjusting batch effects in microarray expression data using empirical Bayes methods, Biostatistics 8 (2007) 118–127.
[23] S. Varma, R. Simon, Bias in error estimation when using cross-validation for model selection, BMC Bioinforma. 7 (2006) 91.
[24] S.A. Miller, D.D. Dykes, H.F. Polesky, A simple salting out procedure for extracting DNA from human nucleated cells, Nucleic Acids Res. 16 (1988) 1215.
[25] L. Breiman, Random Forests, Mach. Learn. 45 (2001) 5–32.
[26] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, et al., Scikit-learn: machine learning in python, J. Mach. Learn. Res. 12 (2011) 2825–2830.
[27] T. Hastie, R. Tibshirani, J. Friedman, The Elements of Statistical Learning, Springer New York Inc., New York, NY, USA, 2001, pp. 592–593.
[28] C. Davidson-Pilon, Lifelines, GitHub Repository, 2015. https://github.com/CamDavidsonPilon/lifelines.
[29] E. Jones, E. Oliphant, P. Peterson, et al., SciPy: Open Source Scientific Tools for Python, 2001. http://www.scipy.org/.
[30] Cancer Genome Atlas Network, Comprehensive molecular characterization of human colon and rectal cancer, Nature 487 (2012) 330–337.
[31] K.A. Heichman, Blood-based testing for colorectal cancer screening, Mol. Diagn. Ther. 18 (2014) 127–135.
[32] M. Valladares-Ayerbes, M. Blanco-Calvo, M. Reboredo, M.J. Lorenzo-Patiño, P. Iglesias-Díaz, M. Haz, et al., Evaluation of the adenocarcinoma-associated gene AGR2 and the intestinal stem cell marker LGR5 as biomarkers in colorectal cancer, Int. J. Mol. Sci. 13 (2012) 4367–4387.

[33] P. Dalerba, D. Sahoo, S. Paik, X. Guo, G. Yothers, N. Song, et al., CDX2 as a prognostic biomarker in stage II and stage III colon cancer, N. Engl. J. Med. 374 (2016) 211–222.
[34] B. Mlecnik, G. Bindea, A. Kirilovsky, H.K. Angell, A.C. Obenauf, M. Tosolini, et al., The tumor microenvironment and Immunoscore are critical determinants of dissemination to distant metastasis, Sci. Transl. Med. 8 (2016) 327ra326.
[35] U. Bork, N.N. Rahbari, S. Schölch, C. Reissfelder, C. Kahlert, M.W. Büchler, et al., Circulating tumour cells and outcome in non-metastatic colorectal cancer: a prospective study, Br. J. Cancer 112 (2015) 1306–1313.
[36] J. Tie, I. Kinde, Y. Wang, H. Wong, I. Skinner, R. Wong, et al., Circulating tumor DNA (ctDNA) as a marker of recurrence risk in stage II colon cancer (CC), in: 2014 ASCO Annual Meeting, 2014.